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Physical examination tests of the shoulder: a systematic review with meta-analysis of individual tests

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CRD summary

This review assessed diagnostic **accuracy** of orthopaedic physical examination tests of the shoulder. Data for a range of tests and pathologies were reported; studies were often small and of poor quality, and results were inconsistent. The conclusion that there was a lack of clarity around the clinical usefulness of the tests reflects the weaknesses in the underlying data.

Authors' objectives

To assess diagnostic **accuracy** of individual orthopaedic physical examination tests of the shoulder to provide clinicians with information to determine whether these tests were appropriate for clinical practice.

Searching

MEDLINE, CINAHL and SPORTDiscus databases were searched (1966 to October 2006). Searches included subject specific terms and terms for diagnostic **accuracy** studies. Personal files were handsearched for publications, posters or abstracts. Bibliographies of review articles were screened for additional studies. Only English-language studies were included.

Study selection

Studies that assessed at least one orthopaedic special test (OST) and used surgery, magnetic resonance imaging (MRI) or injection (acromioclavicular joint only) as the reference standard to confirm diagnosis were eligible for inclusion. Included studies were required to report sensitivity and/or specificity (or sufficient data to calculate these). Studies were excluded if the OST was performed under anaesthesia or in cadavers, or if a group of OSTs were evaluated together as a composite physical examination.

The review assessed a wide range of OSTs for various target conditions: impingement; rotator cuff integrity; glenoid labrum integrity; anterior instability; and acromioclavicular joint pathology.

Studies were independently assessed for inclusion by at least two reviewers. Disagreements were resolved by consensus or discussion with a third reviewer.

Assessment of study quality

Methodological quality of included studies was assessed using the 14-item QUADAS tool to assess aspects of reporting quality, use of an adequate reference standard, appropriateness of the participant spectrum, blinding of interpreters of the index test and reference standard, avoidance of verification biases, avoidance of disease progression bias and handling of withdrawals and

uninterpretable results. QUADAS items were scored as yes, no or unclear. The authors defined high-quality studies as those with an overall quality score of at least 10 unequivocal yes items.

One reviewer assessed validity.

Data extraction

Data were extracted on sensitivity and specificity and positive and negative likelihood ratios for each study, test and target condition.

The authors did not state how data were extracted for the review.

Methods of synthesis

Only studies that reported data for both sensitivity and specificity were included in the meta-analyses. Data were available for Neer test for impingement, Hawkins-Kennedy test for impingement and Speed test for labral pathology.

Pooled estimates of sensitivity and specificity were calculated using dr-ROC software and a fixed-effect model (random-effects and fixed-effect models gave similar results) weighted by inverse variance, and 95% confidence intervals (CIs) calculated. Summary diagnostic odds ratio and area under the summary receiver operating characteristic (SROC) curve were calculated.

Between-study heterogeneity was assessed using Cochran's Q and I^2 tests.

Results of the review

Forty-five studies were included in the review. Twenty two of these met the authors' criteria for high quality, but only two were considered to have adequate sample size (number of participants ranged from 12 to 585).

Impingement: Six studies assessed OSTs for impingement. Meta-analysis gave pooled sensitivity and specificity values for the Neer test of 0.79 (95% CI 0.75 to 0.82; four studies) for sensitivity and 0.53 (95% CI 0.48 to 0.58; four studies) for specificity. Pooled estimate of sensitivity for Hawkins-Kennedy test was 0.79 (95% CI 0.75 to 0.82; four studies) and the pooled estimate of specificity was 0.59 (95% CI 0.53 to 0.64; four studies). For both tests, the pooled diagnostic odds ratio had confidence intervals that crossed one. For other tests of impingement, reported sensitivities ranged from 4% to 88% and reported specificities ranged from 28% to 100%.

Rotator cuff integrity: Fifteen studies assessed 20 OSTs for rotator cuff integrity. None of the 10 OSTs for rotator cuff pathology that were examined in more than one study had consistent diagnostic performance. Reported sensitivity values ranged from 10% to 98% and reported specificity values ranged from 9% to 98%. Data for tests not included in meta-analyses suggested that hornblowers's **sign** and external rotation **lag sign** may be useful for tears of the rotator cuff.

Glenoid labrum integrity: Twenty one studies assessed OSTs for labral pathologies. Meta-analysis gave a pooled estimate for sensitivity of Speed test for superior labral anterior to posterior (SLAP) lesions of 0.32 (95% CI 0.24 to 0.42; five studies) and a pooled estimate of specificity of 0.61 (95% CI 0.54 to 0.68; five studies). Pooled diagnostic odds ratio had confidence intervals that crossed one. Other OSTs for labral pathologies reported sensitivities that ranged from 5% to 99% and specificities that ranged from 11% to 98%. Data for tests not included in meta-analyses suggested that biceps load II may be useful for SLAP lesions.

Five studies assessed nine OSTs for various aspects of instability. Reported sensitivity values

ranged from 28% to 100% and reported specificity values ranged from 44% to 99%. Data for tests not included in meta-analyses suggested that apprehension, relocation and anterior release may be useful tests for anterior instability.

Acromioclavicular joint pathology: Three studies assessed 10 OSTs for acromioclavicular joint pathology. Reported sensitivity values ranged from 16% to 96% and reported specificity values ranged from 10% to 96%.

Authors' conclusions

Based on the results of meta-analyses, diagnostic **accuracy** of Neer test for impingement, Hawkins-Kennedy test for impingement and Speed test for labral pathology was limited. Available information showed a lack of clarity with regard to whether common OSTs used in clinical examination were useful in differentially diagnosing pathologies of the shoulder.

CRD commentary

The review addressed a clearly stated research question defined by appropriate inclusion criteria. A range of sources were searched for relevant studies, including unpublished studies. However, the restriction to English-language studies left the review open to potential language bias. The search strategies used for bibliographic databases included methodological terms for diagnostic **accuracy** studies; use of these terms has been shown to reduce the sensitivity of searches and may result in the loss of relevant studies. Measures were taken to reduce error/bias in the inclusion screening process; it was unclear whether similar measures were applied throughout the review process. Methodological quality of included studies was asessed and incorporated in the description of results. The review reported data for a wide range of tests and pathologies. Studies were often small and of poor quality, and results were inconsistent. These weaknesses in the underlying data are reflected in the authors' cautious conclusions.

Implications of the review for practice and research

Practice: The authors stated that there were very few OSTs of the shoulder that appeared to be diagnostically discriminatory and so useful in the clinic. Either supraspinatus/empty can or infraspinatus test may confirm impingement. For rotator cuff tears, hornblower's **sign** may be diagnostic of severe degeneration or absence of the teres minor muscle, external rotation **lag sign** (ERLS) may be diagnostic of an infraspinatus muscle tear and the bear-hug and belly press tests may be useful for ruling in a subscapularis muscle tear. Two tests may have value as tests for any rotator cuff tear: ERLS as a specific confimatory test and supine impingement test as a screening test. Biceps load II test may be diagnostic for SLAP lesions. Apprehension, relocation and anterior release tests all appeared to be diagnostic for anterior instability, especially when apprehension was used as a positive test instead of pain. For acromioclavicular joint pathology, pain with palpation may be useful as a screen and active compression test may have value as a confirmatory test. Overall, these recommendations should be viewed as a guide.

Research: The authors stated that there was a need for large prospective well-designed studies that examined diagnostic **accuracy** of the numerous physical examination tests of the shoulder.

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Bibliographic details

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